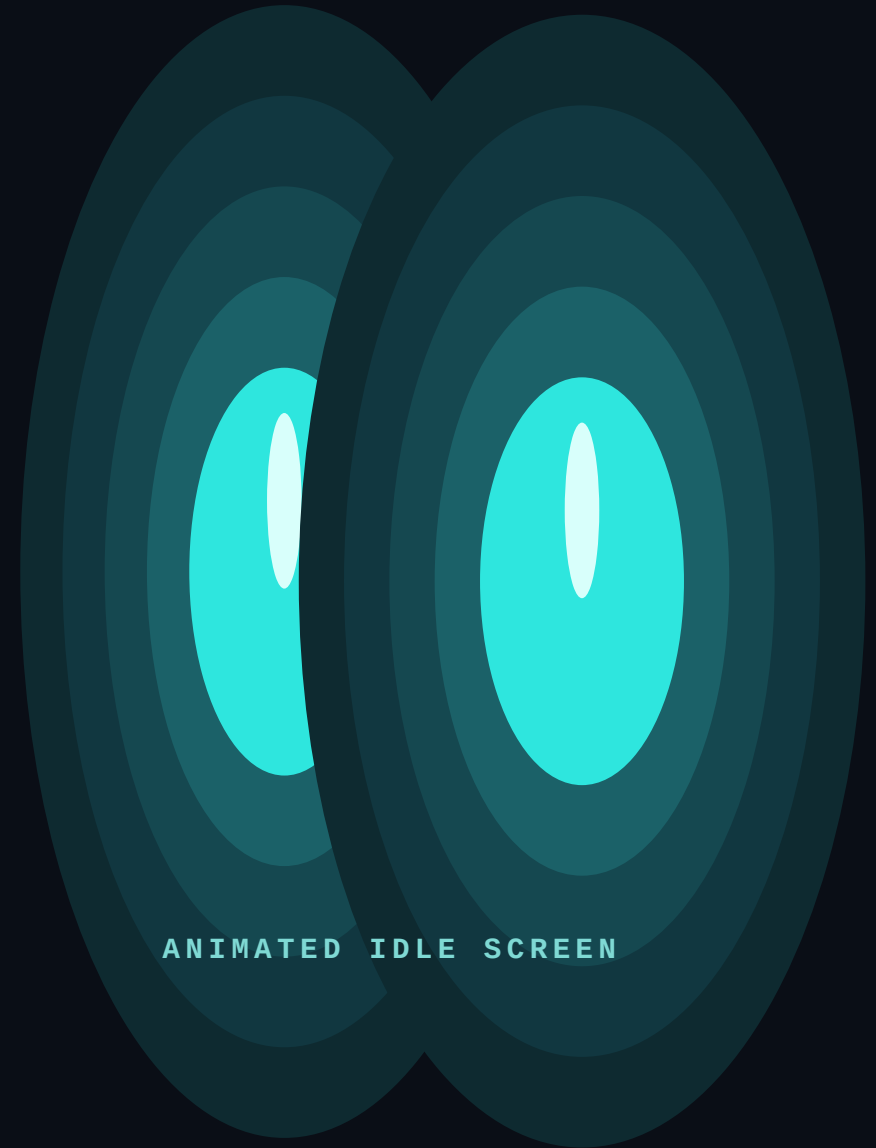


RASPBERRY PI 5 · ON-DEVICE · OFFLINE

Face-Recognition Attendance Kiosk

An offline check-in station that recognizes faces, greets each person by name, and logs attendance to a daily sheet — running entirely on a Raspberry Pi.

python · mediapipe · dlib · pygame · opencv



ANIMATED IDLE SCREEN

What it does

The kiosk turns a Pi, a camera and a screen into a self-serve attendance station.

When someone steps up, its animated eyes lock on, it recognizes them from a photo database, shows a personalized greeting, and records the check-in.

No cards. No app. No internet.

100%

On-device & offline

30

FPS fluid animated UI

4x/s

Recognition checks

1/day

Logged per person

Why face recognition?

Manual register

pen & paper

Roll call eats lesson time, any friend can answer for an absentee, and sheets must be re-typed later.

× slow · proxy-friendly

RFID / ID card

tap a token

Authenticates the card, not the person — cards get borrowed, shared and lost.

× verifies the token

Fingerprint

contact biometric

Verifies the person, but everyone touches one shared sensor and queues build at the door.

× contact · slow queues

Face recognition

this project

Nothing to carry or remember, hard to fake casually — a glance at the camera marks you present.

✓ contactless · on-device

The face is the one credential that can't be forgotten, shared or lost — and it needs no contact at all.

Designed to feel alive

- **Eyes that follow you**

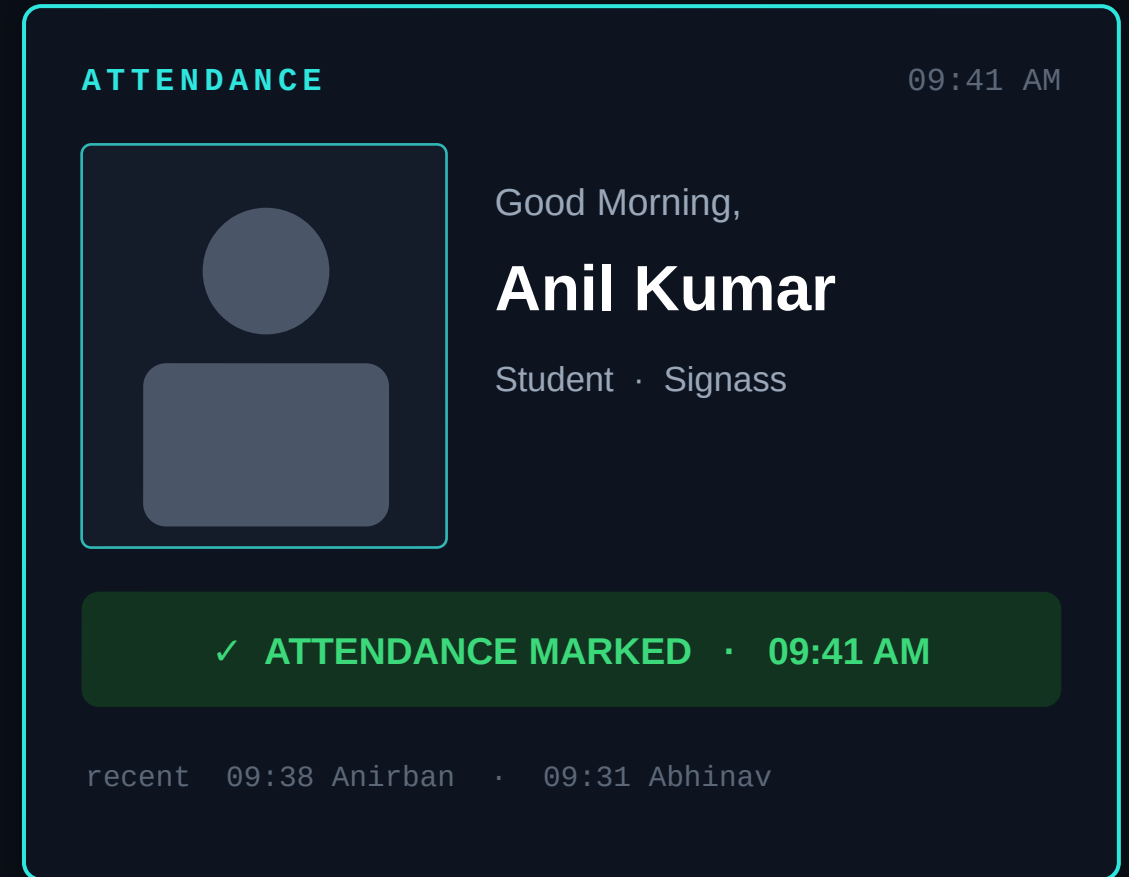
MediaPipe tracks your face; the pupils and eyelids react in real time.

- **Instant, personal greeting**

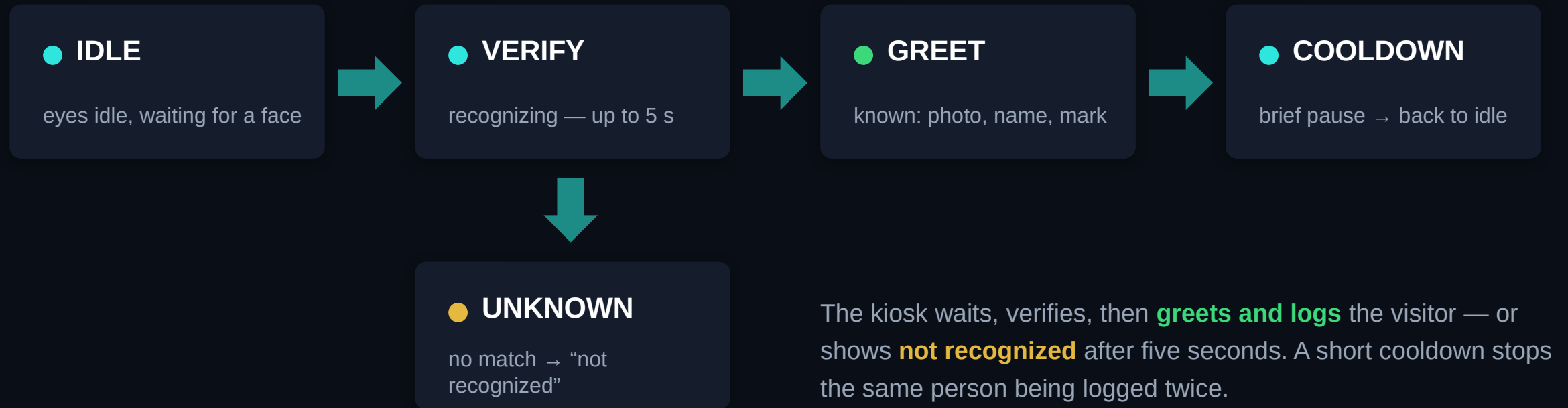
“Good Morning, <name>” with your photo and role the moment you're recognized.

- **Live at a glance**

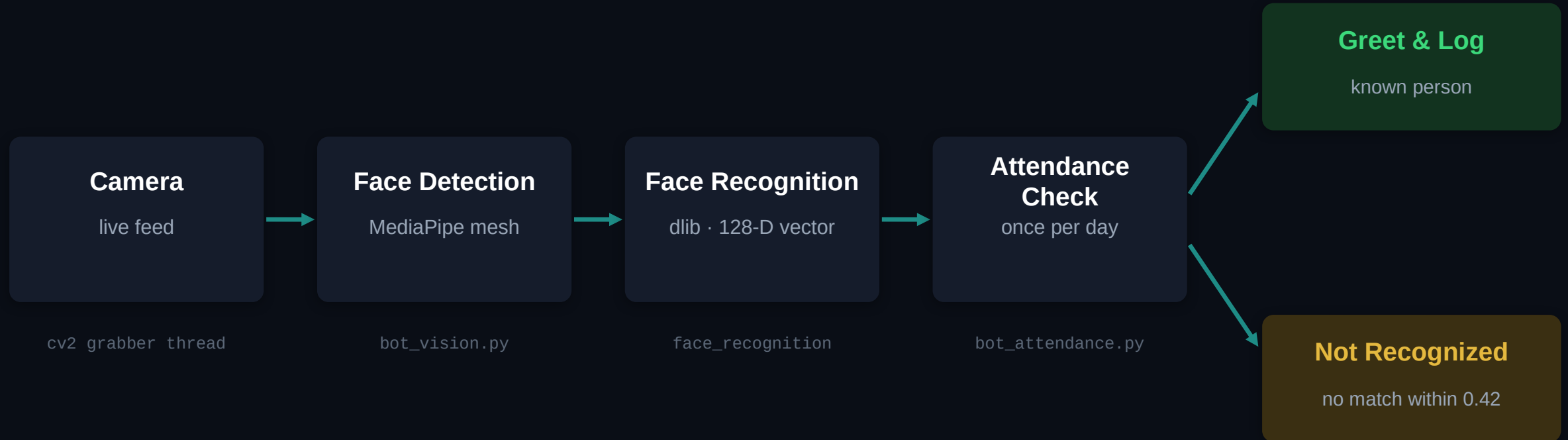
Today's headcount and the last few arrivals sit on the idle screen.



The check-in loop



How the data flows



Every stage runs on the Raspberry Pi itself — no network, no cloud. Recognition happens on a **background thread**, and both outcomes are drawn to the fullscreen pygame display by the UI loop.

From face to record, in four steps

01

DETECT

MediaPipe FaceMesh

468-point mesh finds and frames the face on a 320×240 preview.

02

ENCODE

dlib ResNet

The face crop becomes a 128-number vector — its numeric fingerprint.

03

MATCH

Nearest neighbour

Compared to every known vector; a distance under 0.42 is a match.

04

LOG

Daily CSV

Matched name is written once per person to today's attendance sheet.

Built from open-source parts



Python 3.11

Core runtime & glue



Raspberry Pi 5

Edge hardware — camera + touchscreen



MediaPipe

Face detection & mesh tracking



face_recognition

dlib 128-D face matching



pygame

Fullscreen animated UI



OpenCV

Camera capture & image ops

One clean sheet per day

Name	Category	Time
Anirudha Sarkar	Signass Student	07:30 AM
Aditya Talukdar	Students	07:42 AM
Sharanya Saha	Students	07:55 AM
Rohan Bose	Students	08:10 AM
Sukanya Das	Students	08:20 AM
Bipasa Paul	Students	08:27 AM

attendance/2026-07-17.csv

- **Marked once per day**

A second sighting shows “already marked” — never a duplicate row.

- **Survives restarts**

On launch it reloads today's file and picks up exactly where it left off.

- **Rolls over at midnight**

A fresh sheet each day; the group folder is saved as the category.

Engineered to stay smooth on a Pi

BEFORE

- Recognition ran inline — froze the UI $\sim\frac{1}{2}$ s each try
- Camera reads blocked the main loop
- Face mesh ran on the full 640×480 frame
- Only one recognition attempt every 0.6 s

AFTER

- Recognition on a background worker — UI never stalls
- Dedicated capture thread always holds the latest frame
- Mesh runs on a 320×240 frame ($\sim 4\times$ lighter)
- Four attempts a second, frame-rate capped at 30

2.4×

faster recognition
cadence

3

concurrent threads:
capture · match · UI

0

UI freezes during a scan

Three threads, zero freezes

THREAD 1 — CAMERA GRABBER

Reads the camera continuously and keeps only the newest frame — the rest of the app never waits on camera I/O.



latest frame · lock-protected handoff

THREAD 2 — RECOGNITION WORKER

Every 0.25 s: detect → encode → match. All dlib calls sit behind a single lock — its C++ core is not thread-safe.



match result · polled, non-blocking

THREAD 3 — UI / MAIN LOOP

The pygame state machine draws the eyes and screens at a steady 30 FPS, whatever recognition is doing.

30 FPS

steady UI, always

4x/s

recognition attempts

0

freezes during a scan

More than a doorbell

Live enrollment

Capture a new student at the kiosk — saved and matched instantly.

Hot reload

Rescan the face database in the background, no restart.

Idle stats

Today's headcount and recent arrivals on the home screen.

Face tracking

Cyan eyes follow the visitor's face in real time.

Group filtering

Loads only chosen folders — Signass Student, Students.

Debug view

Live camera, face mesh and an FPS counter overlay.

Proven with real students

TEST SCENARIOS

TC-01	Enrolled student recognized & marked	PASS
TC-02	Re-appearance → “already marked”, no duplicate	PASS
TC-03	Unknown face rejected after 5 s	PASS
TC-04	Kiosk enrollment (E) — matched on next attempt	PASS
TC-05	Folder reload (R) picks up new photos	PASS
TC-06	Phone photos with EXIF rotation	PASS
TC-07	Midnight rollover starts a fresh sheet	PASS

CASE STUDY

The sideways-photo bug

New photos taken on a phone loaded with zero faces detected. Phones store the image sideways and rely on an EXIF tag to rotate it — the loader ignored the tag, so the detector saw a rotated face and gave up.

The fix: apply the EXIF transpose before detection, and downscale 12 MP photos to 800 px first.

0 → 3

photos detected, of 3

71 → 9 s

database reload (8×)

Modules & what's next

ARCHITECTURE

- **main.py**
State machine & orchestration
- **bot_vision.py**
Camera, tracking, recognition, enrollment
- **bot_eyes.py**
pygame UI — eyes, greeting, overlays
- **bot_attendance.py**
Daily CSV logging & de-duplication
- **config.py**
Tunable settings & colour theme

ROADMAP

- Export / sync attendance to Google Sheets
- Multi-face handling for busy entrances
- Liveness check to resist photo spoofing
- Lightweight web dashboard for monitoring

On-device attendance – eyes and all.

